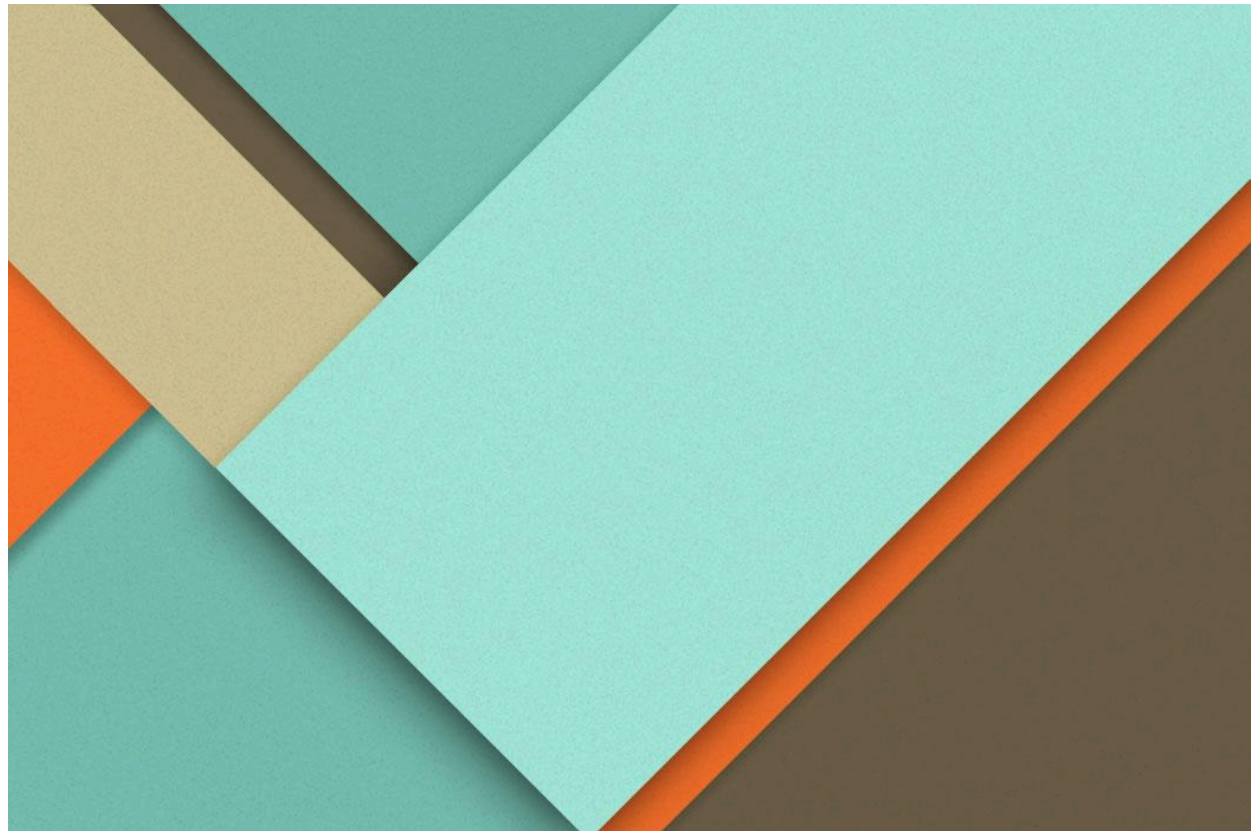




DNS Resolver & Query Analyzer with Performance Comparison

CS3001 – Computer Networks : Project Proposal

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Overview

The Domain Name System (DNS) is a critical component of the Internet that translates human-readable domain names into machine-readable IP addresses. When users access websites such as google.com or facebook.com, DNS servers are responsible for resolving the domain name into an IP address so that communication between the client and server can be established.

The objective of this project is to develop a DNS Resolver and Query Analyzer using Python. The system will allow users to send DNS queries to public DNS servers and analyze the responses returned by those servers. The program will demonstrate how DNS resolution works and how different DNS servers respond to queries.

The resolver will be implemented using the Python dnspython library, which allows sending DNS queries and receiving responses programmatically. Users will be able to input a domain name and select the type of DNS record they want to retrieve. The system will then query DNS servers and display the results, including the returned IP address, response time, and other relevant DNS information.

To further analyze DNS performance, the system will query multiple DNS servers and compare their response times. Public DNS servers such as Google DNS (8.8.8.8), Cloudflare DNS (1.1.1.1), and OpenDNS (208.67.222.222) will be used to evaluate performance differences.

The project will also include DNS packet analysis using Wireshark, which will capture and analyze DNS query and response packets generated by the resolver. This will allow visualization of DNS traffic and help demonstrate how DNS communication occurs between clients and DNS servers.

Additionally, the system will store DNS query results in a log file for later analysis. This logging feature will help track query performance and provide useful data for evaluating DNS response behavior.



Overall, this project aims to provide a practical implementation of DNS resolution while demonstrating important networking concepts such as client-server communication, network latency, and packet-level analysis.

Functional Features

The proposed system will include the following functional features:

I. Domain Name Resolution

The system will allow users to enter a domain name and retrieve the corresponding IP address using DNS queries.

II. DNS Record Type Selection

Users will be able to choose the type of DNS record they want to query. Supported record types include:

- A Record (IPv4 address)
- AAAA Record (IPv6 address)
- MX Record (mail servers)
- NS Record (name servers)
- CNAME Record (canonical name)

III. DNS Server Comparison

The resolver will query multiple public DNS servers and compare their response times to determine which server responds fastest.

DNS servers used:

- Google DNS (8.8.8.8)
- Cloudflare DNS (1.1.1.1)
- OpenDNS (208.67.222.222)

IV. DNS Response Time Measurement

The system will calculate and display the time required to receive a response from the DNS server.

V. TTL (Time-To-Live) Display

The resolver will extract and display the TTL value returned by DNS records.

VI. DNS Query Logging

All DNS query results will be saved to a log file containing:

- Domain name
- Record type
- DNS server used
- Response time
- Result returned
- Timestamp of the query

VII. Packet Capture and Analysis

DNS packets generated by the resolver will be captured using Wireshark to analyze the DNS communication process.

VIII. Error Handling

The system will handle invalid domain names and network errors gracefully and notify the user accordingly.

Plan of Work

Since this project will be completed individually, all tasks will be organized into weekly milestones.

Week 1 – Research and Environment Setup

Tasks:

- Study DNS protocol and DNS record types
- Understand DNS query and response process
- Install and configure required tools:
 - Python
 - dnspython library
 - Wireshark
- Review DNS packet structure and DNS headers

Week 2 – Basic DNS Resolver Development

Tasks:

- Develop basic DNS resolver in Python
- Allow user to input domain name
- Retrieve A record (IPv4 address)
- Display results in terminal output

Week 3 – Multi-Record Query Support

Tasks:

- Implement support for multiple DNS record types:
 - MX
 - NS
 - CNAME
 - AAAA
- Display TTL information
- Implement record type selection menu

Week 4 – Performance Analysis Features

Tasks:

- Implement response time measurement
- Query multiple DNS servers
- Compare response times between servers

- Identify fastest responding DNS server

Week 5 – Logging, Packet Analysis, and Final Testing

Tasks:

- Implement query logging system
- Save results to a log file
- Capture DNS packets using Wireshark
- Analyze DNS queries and responses
- Perform testing and debugging
- Prepare final report and presentation

References

1. Mockapetris, P. (1987). **Domain Names – Implementation and Specification (RFC 1035)**.
2. Kurose, J., & Ross, K. **Computer Networking: A Top-Down Approach**.
3. dnspython Documentation – <https://www.dnspython.org>
4. Wireshark Network Protocol Analyzer – <https://www.wireshark.org>